

PROFILE

MPhil researcher at the University of Sydney building reliable, efficient AI systems across federated learning, semantic communication, and trustworthy machine learning.

CURRENT RESEARCH AND TEACHING

Casual Academic

UNSW Sydney | Aug 2026 - present

Teaching and academic support in computing coursework.

- Facilitate classes and support students through technical concepts and practical problem solving.
- Apply rubric-aligned feedback and clear explanations to student work.

Casual Academic Tutor

The University of Sydney | Feb 2026 - present

Tutorial delivery and student learning support for university coursework.

- Deliver weekly tutorials, class activities, small-group discussions, and question sessions.
- Support academic communication, teamwork, technical problem solving, and professional practice.
- Coordinate with unit teams to deliver consistent, inclusive learning experiences.

Research Assistant

DUAL Group, The University of Sydney | Apr 2024 - present

Research engineering and experimentation for distributed learning and efficient AI systems.

- Build reproducible training and evaluation pipelines with explicit configs, seeds, logging, and reporting.
- Implement baselines, targeted ablations, and publication-quality figures and tables.
- Coordinate experiment ownership and artifact handoff across code, plots, and manuscripts.

SELECTED RESEARCH

Distributionally Robust Wireless Semantic Communication with Large AI Models

2025-2026 | published

Contributed modelling, experiment design, robust training simulations, and evaluation under constrained channels. Evidence: JSAC 2026 publication venue.

Federated Deep Equilibrium Learning over Resource-Constrained Edge Networks

2025-present | submitted

Implemented FeDEQ components, explored communication-efficient aggregation, and evaluated non-IID NLP and vision settings. Evidence: NLP + vision evaluation domains.

BRAVE: Block-wise Structural Regularization via Controlled Evidence Feedback for Reliable Label Aggregation under Sparse Crowdsourcing

2026 | under review

Zerun Niu - first author; led algorithm design, literature review, experimental design, code implementation, and experimental deployment. Evidence: 14 crowdsourcing benchmarks; 5/14 lowest NLL; 9/14 best or tied-best ECE; 11/14 within 0.03 accuracy of the strongest baseline.

RESEARCH ENGINEERING CASEWORK

DUAL Group Research Website

2024-present | active

Designed, developed, deployed, and maintain the group's official web presence and research communication system. Evidence: Design -> deploy end-to-end ownership.

BRAVE - first-author evidence

Under review at TMLR

Designed controlled evidence feedback for sparse crowdsourcing and led the complete research pipeline.

- 14 benchmarks; lowest NLL on 5/14; best or tied-best ECE on 9/14.
- Accuracy within 0.03 of the strongest external baseline on 11/14.
- Completed downstream reward-model calibration transfer experiments.

TECHNICAL PRACTICE

Research methods	Algorithm design · Literature synthesis · Experimental design · Ablation studies · Calibration · Robustness evaluation
ML engineering	Python · PyTorch · Hugging Face · scikit-learn · Reproducible pipelines · Experiment tracking
Distributed systems	Federated learning · Ray · Dask · Spark · Airflow · FastAPI
GenAI	LLM and VLM orchestration · RAG · Agents · LoRA and PEFT · Evaluation · Safety guardrails
Web and research tooling	TypeScript · React · Astro · GitHub Actions · Data visualisation · Research communication

EDUCATION

Master of Philosophy in Computer Science

The University of Sydney | 2025 - present

Research in reliable and efficient AI systems, supervised within the DUAL Group.

- Research across federated learning, semantic communication, calibration, and resource-constrained AI.
- Build reproducible methods and experimental pipelines from literature review through analysis.

Bachelor of Advanced Computing, Data Science

The University of Sydney | 2021 - 2025

Graduated with Distinction; final-year work centred on machine learning and distributed systems.

- Undergraduate thesis in adaptive low-rank optimisation for federated learning, marked 85/100.
- Coursework included machine learning, deep learning, natural language processing, and distributed systems.